DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 8

Quantitative researcher

Profession 8 of 10 in Development and IT.

What the profession means

A quantitative researcher designs studies with numbers: surveys, experiments and statistical models that test whether a business idea or policy has a measurable effect.

What professionals do every day

Designs the study, decides the sample size, collects or cleans the data, runs the statistical tests, and writes a report with confidence and limits.

Why businesses hire them, and what problems they solve

Businesses and institutions need to know whether a change worked, not just that it looked good.

- Results that are chance.
- Samples too small to answer the question.
- Reports that overstate certainty.

Skills you need

- Study design and sampling.
- Hypothesis testing and confidence intervals.
- Regression basics.
- Clear reporting of uncertainty.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
Python and pandas	Analysis in code	Free	Free	pandas.pydata.org
Jamovi	Free statistics software	Free	Free	www.jamovi.org
R (open source)	Statistical language	Free	Free	www.r-project.org

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	Compute a mean and a confidence interval.
Intermediate	Design a sample size for a survey.
Advanced	Test a difference between two groups.

Projects by level

- **Beginner:** Customer satisfaction study.
- **Intermediate:** A/B test readout.

- **Advanced:** Market sizing survey.

Client-ready portfolio project

A survey analysis for a fictional firm: sample size, a difference test between two groups and a report with confidence and limits.

Worked example

Worked example. A difference of 5 points between two groups with a wide interval is reported as 'not clear yet', not as a win.

Assignment with full solution

ASSIGNMENT

Difference between two groups

Business problem: A firm wants to know if a new onboarding improves a score.

Steps

1. State the null hypothesis: no difference.
2. Compute the difference in means and its standard error.
3. Report the result with a confidence interval.
4. Write the conclusion in plain words, including the limit.

Expected output: The estimate, interval and conclusion.

SOLUTION (ANSWER KEY)

The conclusion depends on whether the interval excludes zero, and it must say the sample size. A wide interval means the evidence is not yet clear, which is a valid answer.

Common mistakes

- Testing many hypotheses without correction.
- Small samples reported as firm results.
- Confusing correlation with cause.
- Hiding the uncertainty.

Freelancing opportunities

- Survey and experiment analysis for firms.
- Research support for start-ups.
- Briefs for 'statistics', 'survey analysis', 'hypothesis testing'.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- Jamovi learning resources

Part of **Master Freelancing: Data Analytics and Data Science**. **Back to the chapter list on skillnavora.in** · **Open the full textbook**. Official links in this chapter were not checked live; confirm before relying on them.